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DES SCIENCES

Université: Institut Universitaire des Sciences (IUS)

TD N° 8: Réseau1

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Niveau: 3^{ème} Année

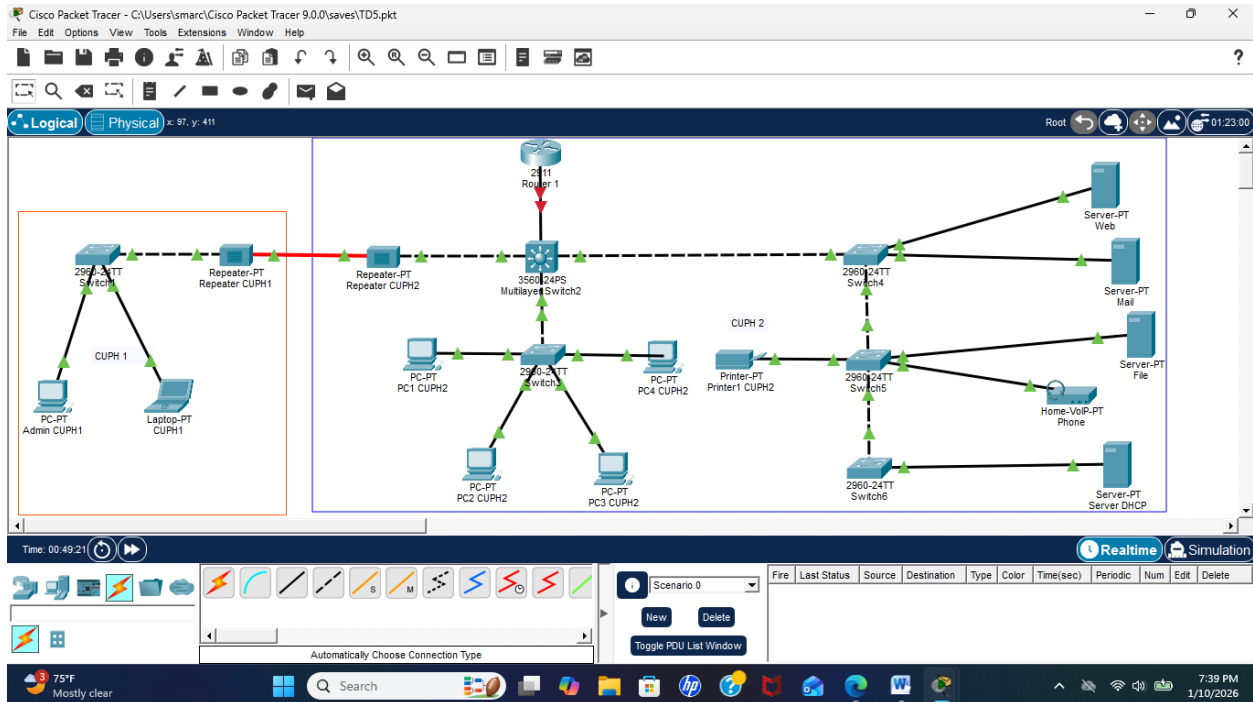
Date: Le11/Janvier/2025

Objectif du TD

L'objectif de ce travail dirigé est de créer et configurer plusieurs VLAN sur des switchs afin de segmenter le réseau. Les ports sont assignés aux VLAN correspondants et un routeur est utilisé pour permettre la communication entre les différents VLAN. Ce TP permet de mieux comprendre la segmentation réseau et le routage inter-VLAN.

1. Reproduisez cette topologie en configurant plusieurs VLAN sur les switches. Assignez les ports aux VLAN correspondants, puis testez et gérez la connectivité entre les eux à l'aide d'un routeur.

a) J'ai Reproduit la topologie



1. Configuration des VLANs sur les switches

a) Pour le Switches Principale

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Ventes
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
```

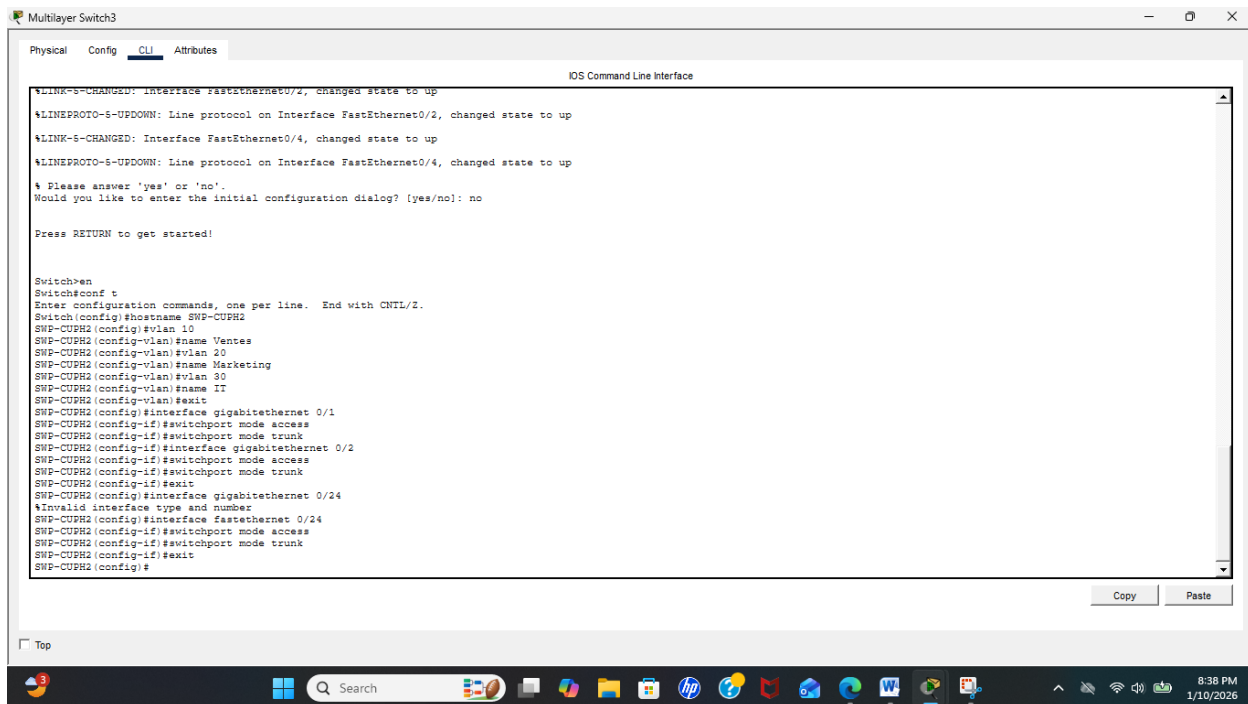
2. Configuration du Switch Principal (Core Switch)

Configuration des ports trunk:

```
Switch(config)# interface gigabitethernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport mode trunk
Switch(config-if)# interface gigabitethernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

3. Configuration du port vers le routeur (Router-on-a-Stick) :

```
Switch(config)# interface FastEthernet 0/24
Switch(config-if)# switchport mode access
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```



The screenshot shows a Windows desktop with a window titled "Multilayer Switch3". The window has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" selected. The main area displays the "IOS Command Line Interface". The text in the CLI window shows the following sequence of events:

- Status messages: `%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up`, `%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up`, `%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up`, and `%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up`.
- Initial configuration dialog prompts: `% Please answer 'yes' or 'no'. Would you like to enter the initial configuration dialog? [yes/no]: no` and `Press RETURN to get started!`.
- User input and commands: `Switch>en`, `Switch#conf t`, `Enter configuration commands, one per line. End with CNTL/Z.`, `Switch(config)#hostname SWP-CUPH2`, `SWP-CUPH2(config)#vlan 10`, `SWP-CUPH2(config-vlan)#name Ventes`, `SWP-CUPH2(config-vlan)#vlan 20`, `SWP-CUPH2(config-vlan)#name Marketing`, `SWP-CUPH2(config-vlan)#vlan 30`, `SWP-CUPH2(config-vlan)#name IT`, `SWP-CUPH2(config-vlan)#exit`, `SWP-CUPH2(config)#interface gigabitethernet 0/1`, `SWP-CUPH2(config-if)#switchport mode access`, `SWP-CUPH2(config-if)#switchport mode trunk`, `SWP-CUPH2(config-if)#interface gigabitethernet 0/2`, `SWP-CUPH2(config-if)#switchport mode access`, `SWP-CUPH2(config-if)#switchport mode trunk`, `SWP-CUPH2(config-if)#exit`, `SWP-CUPH2(config)#interface gigabitethernet 0/24`, `%Invalid interface type and number`, `SWP-CUPH2(config)#interface fastethernet 0/24`, `SWP-CUPH2(config-if)#switchport mode access`, `SWP-CUPH2(config-if)#switchport mode trunk`, `SWP-CUPH2(config-if)#exit`, and `SWP-CUPH2(config)#`.

At the bottom of the CLI window, there are "Copy" and "Paste" buttons. The Windows taskbar at the very bottom shows the time as 8:38 PM on 1/10/2026.

2. Configuration des VLANs sur les switches

b) Pour le Switches1-CUPH1

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Ventes
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Marketing
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name IT
Switch(config-vlan)# exit
```

3. Configuration des autres switches

- **Sur Switch1-CUPH1 et de Switch3-CUPH2 à Switch6-CUPH2:**

Configuration des ports trunk vers le switch principal

```
Switch(config)# interface Fastethernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# exit
```

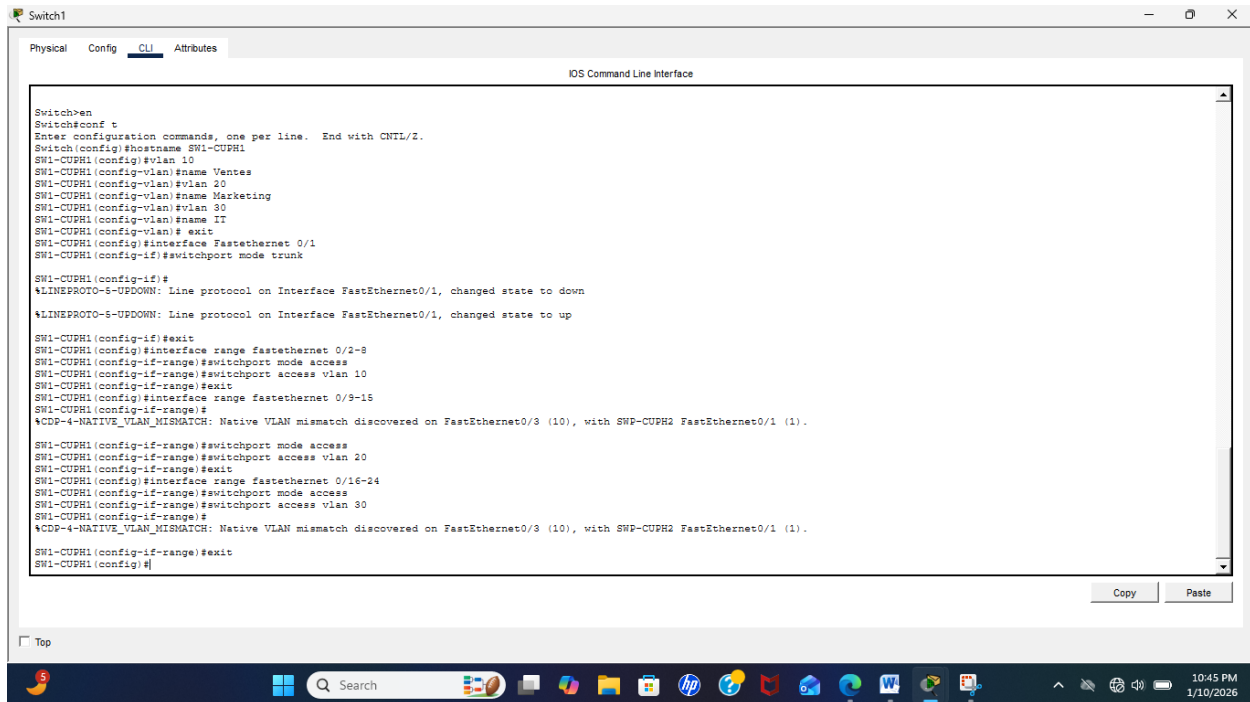
! Attribution des ports aux VLANs (exemple)

```
Switch(config)# interface range fastethernet 0/2-8
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastethernet 0/9-15
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

```
Switch(config)# interface range fastethernet 0/16-24
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
```

Switch(config-if-range)# exit



```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1-CUPH1
SW1-CUPH1(config)#vlan 10
SW1-CUPH1(config-vlan)#name Ventes
SW1-CUPH1(config-vlan)#vlan 20
SW1-CUPH1(config-vlan)#name Marketing
SW1-CUPH1(config-vlan)#vlan 30
SW1-CUPH1(config-vlan)#name IT
SW1-CUPH1(config-vlan)# exit
SW1-CUPH1(config)#interface FastEthernet 0/1
SW1-CUPH1(config-if)#switchport mode trunk

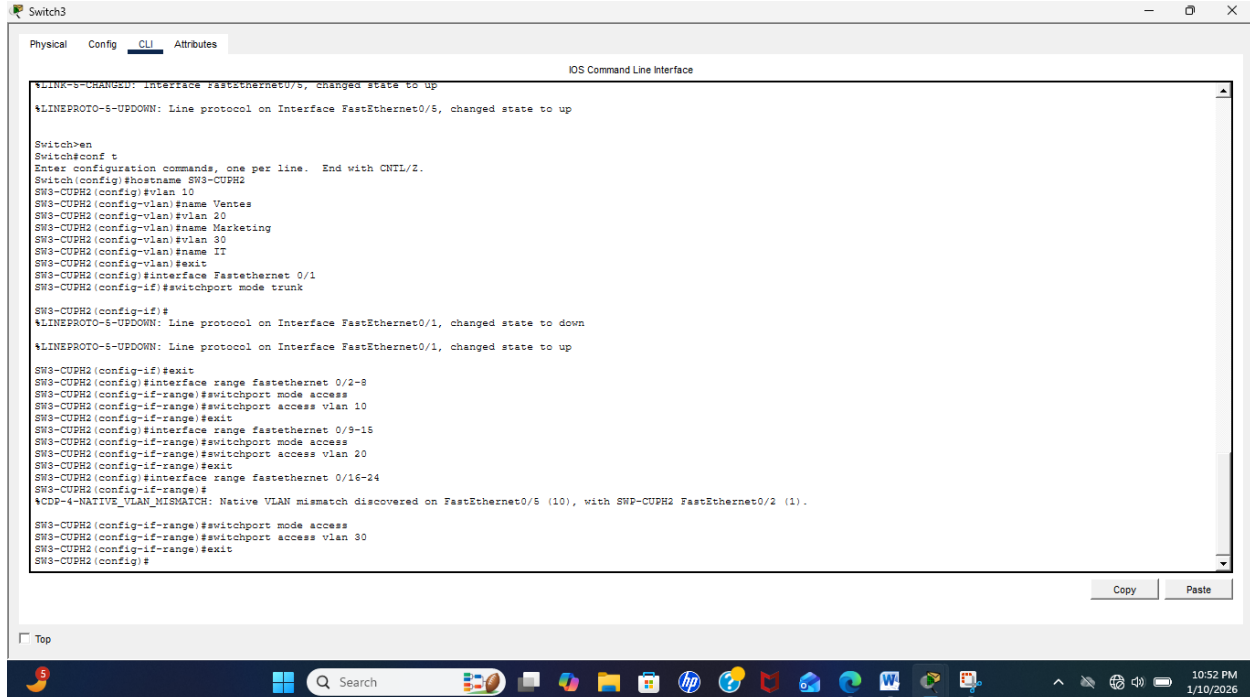
SW1-CUPH1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

SW1-CUPH1(config-if)#exit
SW1-CUPH1(config)#interface range fastethernet 0/2-8
SW1-CUPH1(config-if-range)#switchport mode access
SW1-CUPH1(config-if-range)#switchport access vlan 10
SW1-CUPH1(config-if-range)#exit
SW1-CUPH1(config)#interface range fastethernet 0/9-15
SW1-CUPH1(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/9 (10), with SWP-CUPH2 FastEthernet0/1 (1).

SW1-CUPH1(config-if-range)#switchport mode access
SW1-CUPH1(config-if-range)#switchport access vlan 20
SW1-CUPH1(config-if-range)#exit
SW1-CUPH1(config)#interface range fastethernet 0/16-24
SW1-CUPH1(config-if-range)#switchport mode access
SW1-CUPH1(config-if-range)#switchport access vlan 30
SW1-CUPH1(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (10), with SWP-CUPH2 FastEthernet0/1 (1).

SW1-CUPH1(config-if-range)#exit
SW1-CUPH1(config)#
```

• Sur Switch3-CUPH2



```
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW3-CUPH2
SW3-CUPH2(config)#vlan 10
SW3-CUPH2(config-vlan)#name Ventes
SW3-CUPH2(config-vlan)#vlan 20
SW3-CUPH2(config-vlan)#name Marketing
SW3-CUPH2(config-vlan)#vlan 30
SW3-CUPH2(config-vlan)#name IT
SW3-CUPH2(config-vlan)#exit
SW3-CUPH2(config)#interface FastEthernet 0/1
SW3-CUPH2(config-if)#switchport mode trunk

SW3-CUPH2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

SW3-CUPH2(config-if)#exit
SW3-CUPH2(config)#interface range fastethernet 0/2-8
SW3-CUPH2(config-if-range)#switchport mode access
SW3-CUPH2(config-if-range)#switchport access vlan 10
SW3-CUPH2(config-if-range)#exit
SW3-CUPH2(config)#interface range fastethernet 0/9-15
SW3-CUPH2(config-if-range)#switchport mode access
SW3-CUPH2(config-if-range)#switchport access vlan 20
SW3-CUPH2(config-if-range)#exit
SW3-CUPH2(config)#interface range fastethernet 0/16-24
SW3-CUPH2(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (10), with SWP-CUPH2 FastEthernet0/2 (1).

SW3-CUPH2(config-if-range)#switchport mode access
SW3-CUPH2(config-if-range)#switchport access vlan 30
SW3-CUPH2(config-if-range)#exit
SW3-CUPH2(config)#
```

- **Sur Switch4-CUPH2**

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW4-CUPH2
SW4-CUPH2(config)#vlan 10
SW4-CUPH2(config-vlan)#name Ventas
SW4-CUPH2(config-vlan)#vlan 20
SW4-CUPH2(config-vlan)#name Marketing
SW4-CUPH2(config-vlan)#vlan 30
SW4-CUPH2(config-vlan)#name IT
SW4-CUPH2(config-vlan)#exit
SW4-CUPH2(config)#interface FastEthernet 0/1
SW4-CUPH2(config-if)#switchport mode trunk

SW4-CUPH2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

SW4-CUPH2(config-if)#exit
SW4-CUPH2(config)#interface range fastEthernet 0/2-8
SW4-CUPH2(config-if-range)#switchport mode access
SW4-CUPH2(config-if-range)#switchport access vlan 10
SW4-CUPH2(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (10), with Switch FastEthernet0/1 (1).
SW4-CUPH2(config-if-range)#exit
SW4-CUPH2(config)#interface range fastEthernet 0/9-15
SW4-CUPH2(config-if-range)#switchport mode access
SW4-CUPH2(config-if-range)#switchport access vlan 20
SW4-CUPH2(config-if-range)#exit
SW4-CUPH2(config)#interface range fastEthernet 0/16-24
SW4-CUPH2(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (10), with Switch FastEthernet0/1 (1).
SW4-CUPH2(config-if-range)#switchport mode access
SW4-CUPH2(config-if-range)#switchport access vlan 30
SW4-CUPH2(config-if-range)#exit
SW4-CUPH2(config)#

```

- **Sur Switch5-CUPH2**

```

SW5-CUPH2(config-vlan)#name Ventas
SW5-CUPH2(config-vlan)#vlan 20
SW5-CUPH2(config-vlan)#name Marketing
SW5-CUPH2(config-vlan)#vlan 30
SW5-CUPH2(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4-CUPH2 FastEthernet0/4 (10).
SW5-CUPH2(config-vlan)#name IT
SW5-CUPH2(config-vlan)#exit
SW5-CUPH2(config)#interface FastEthernet 0/1
SW5-CUPH2(config-if)#switchport mode trunk

SW5-CUPH2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

SW5-CUPH2(config-if)#exit
SW5-CUPH2(config)#interface range fastEthernet 0/2-8
SW5-CUPH2(config-if-range)#switchport mode access
SW5-CUPH2(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4-CUPH2 FastEthernet0/4 (10).
SW5-CUPH2(config-if-range)#switchport access vlan 10
SW5-CUPH2(config-if-range)#exit
SW5-CUPH2(config)#interface range fastEthernet 0/9-15
SW5-CUPH2(config-if-range)#switchport mode access
SW5-CUPH2(config-if-range)#switchport access vlan 20
SW5-CUPH2(config-if-range)#exit
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/5 (10), with Switch FastEthernet0/1 (1).
SW5-CUPH2(config)#interface range fastEthernet 0/16-24
SW5-CUPH2(config-if-range)#switchport mode access
SW5-CUPH2(config-if-range)#switchport access vlan 30
SW5-CUPH2(config-if-range)#exit
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with SW4-CUPH2 FastEthernet0/4 (10).
SW5-CUPH2(config)#exit
SW5-CUPH2#
%SYS-5-CONFIG_I: Configured from console by console

```

- **Sur Switch6-CUPH2**

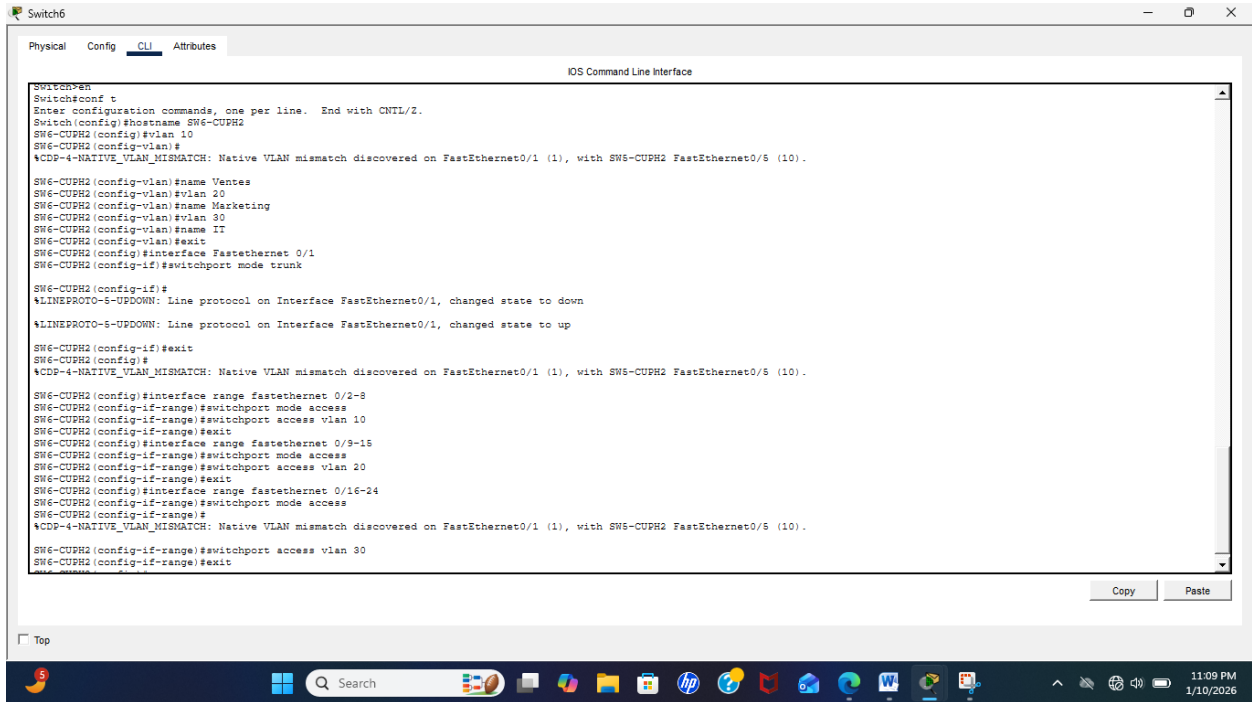


Figure 1 Sur ces images, j'ai configuré les switches (Voir page 6 à 8).

4. Configuration du Routeur (Router-on-a-Stick)

```
Router> enable
```

Router# configure terminal

```
Router(config)# interface gigabitethernet 0/0.10
```

```
Router(config-subif)# encapsulation dot1Q 10
```

```
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
```

```
Router(config-subif)# exit
```

```
Router(config)# interface gigabitethernet 0/0.20
```

```
Router(config-subif)# encapsulation dot1Q 20
```

```
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
```

Router(config-subif)# exit

```
Router(config)# interface gigabitethernet 0/0.30
```

```
Router(config-subif)# encapsulation dot1Q 30
```

```
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

```
Router(config-subif)# exit
```


! Activation de l'interface physique

Router(config)# interface gigabitethernet 0/0

Router(config-if)# no shutdown

Router(config-if)# exit

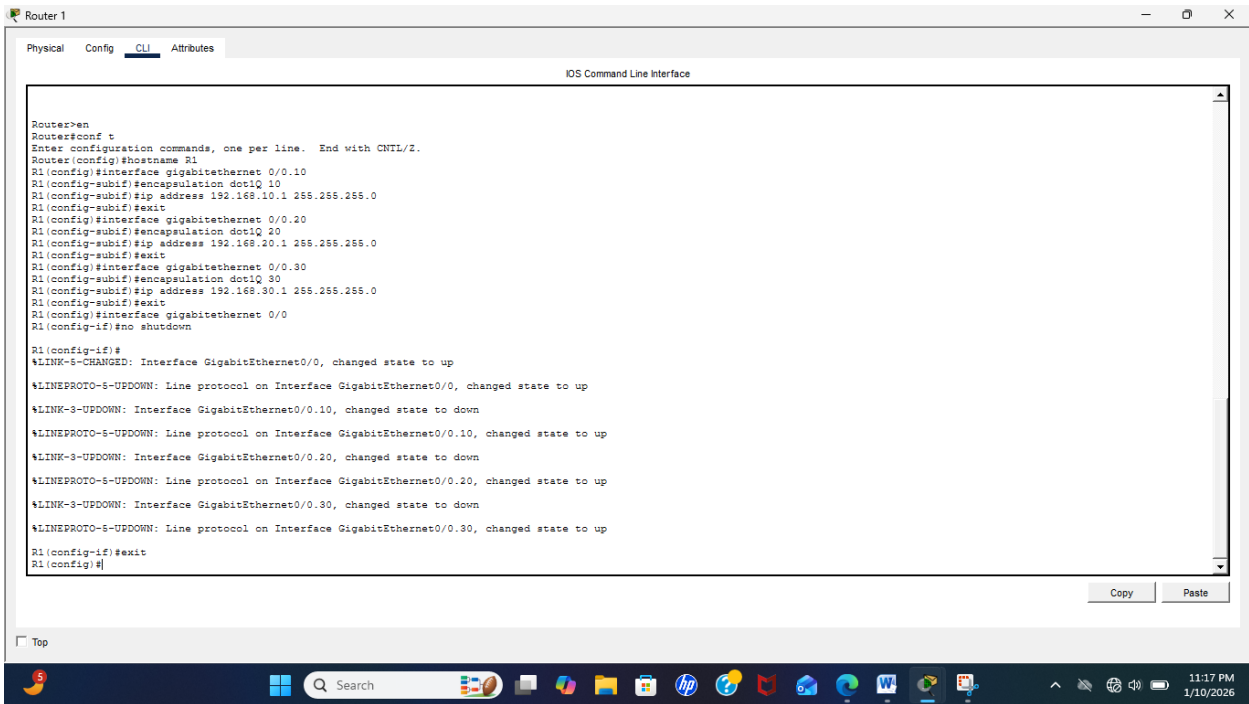


Figure 2 Sur cette image, j'ai configuré l'interface physique du routeur

5. Configuration VTP (pour synchroniser les VLANs)

a) Sur le Switch Principal (VTP Server):

```
Switch(config)# vtp domain cuph.com
```

```
Switch(config)# vtp password 9999
```

```
Switch(config)# vtp mode server
```

```
Switch(config)# vtp version 2
```

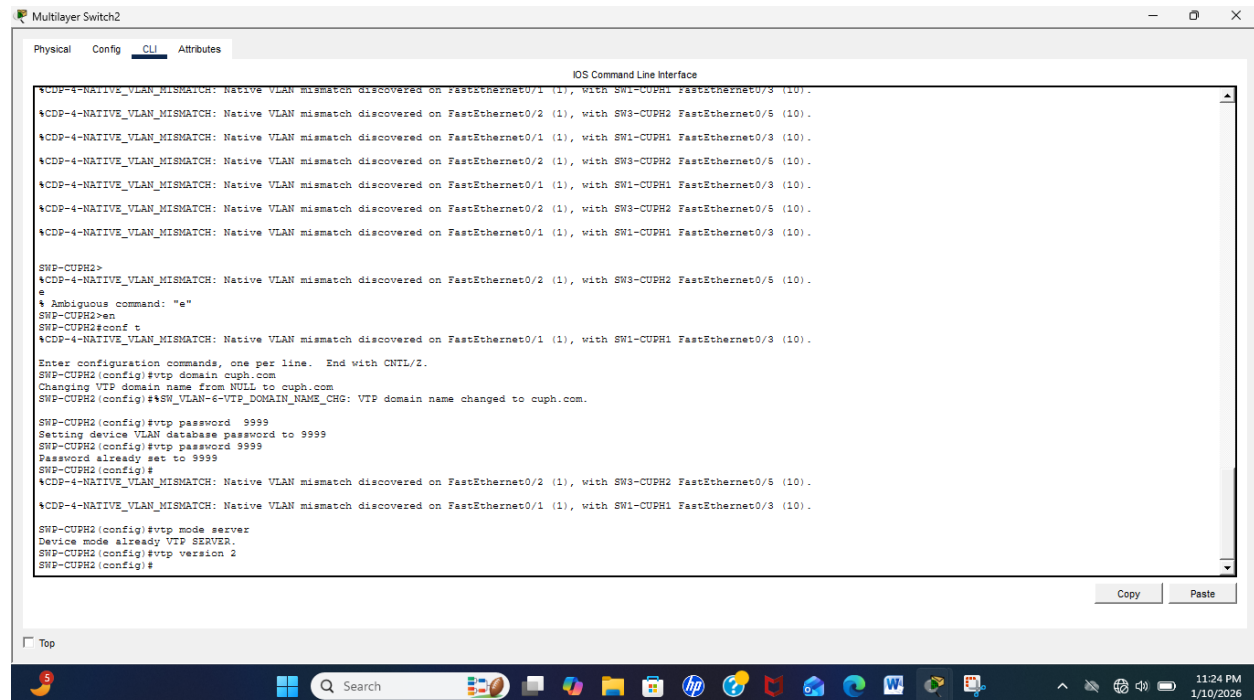


Figure 3 Sur cette image, j'ai configure le Switchs principale.

b) Sur les autres switches (VTP Clients):

```
Switch(config)# vtp domain cuph.com
```

```
Switch(config)# vtp password 9999
```

```
Switch(config)# vtp mode client
```


6. Configuration STP (Spanning Tree)

- Pour éviter les boucles:

Switch(config)# spanning-tree mode rapid-pvst

Switch(config)# spanning-tree vlan 10 root primary

Switch(config)# spanning-tree vlan 20 root secondary

7. Configuration sécurité des ports

Switch(config)# interface fastethernet 0/1

Switch(config-if)# switchport port-security

Switch(config-if)# switchport port-security maximum 3

Switch(config-if)# switchport port-security violation restrict

Switch(config-if)# exit

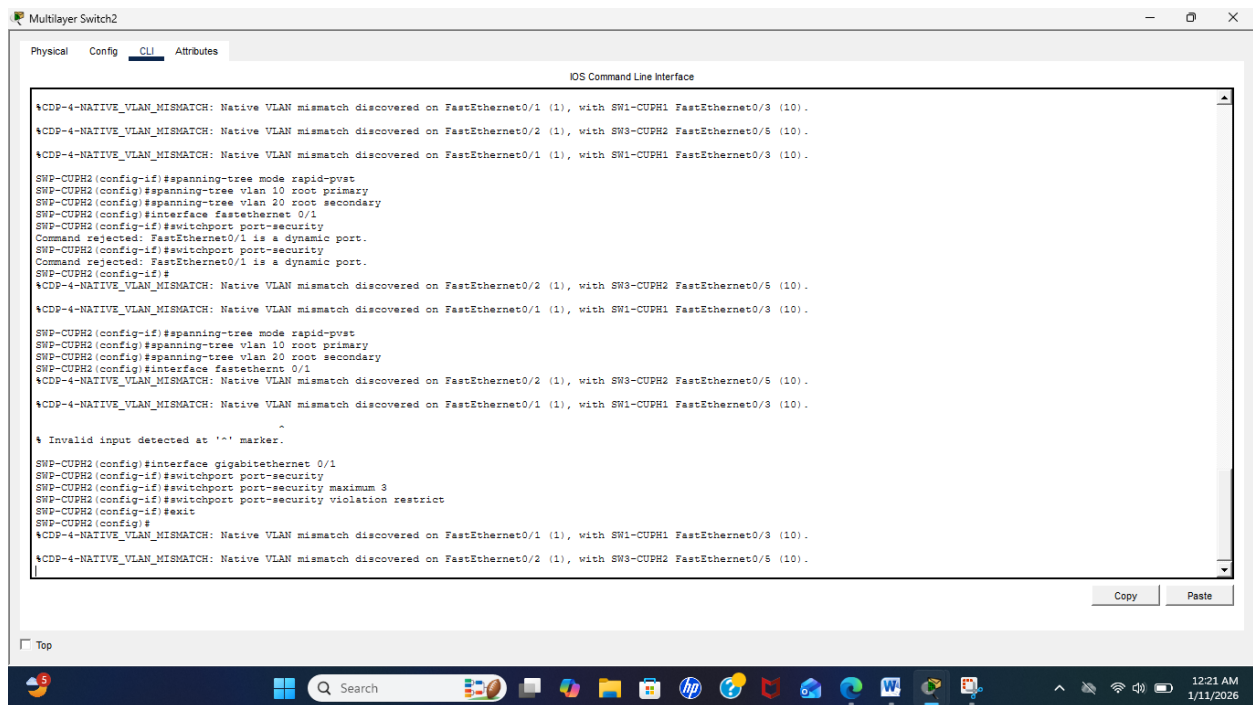


Figure 4 Sur cette image, j'ai configuré le Switches

8. Vérifications importantes:

a) Vérifier les VLANs :

Switch# show vlan brief

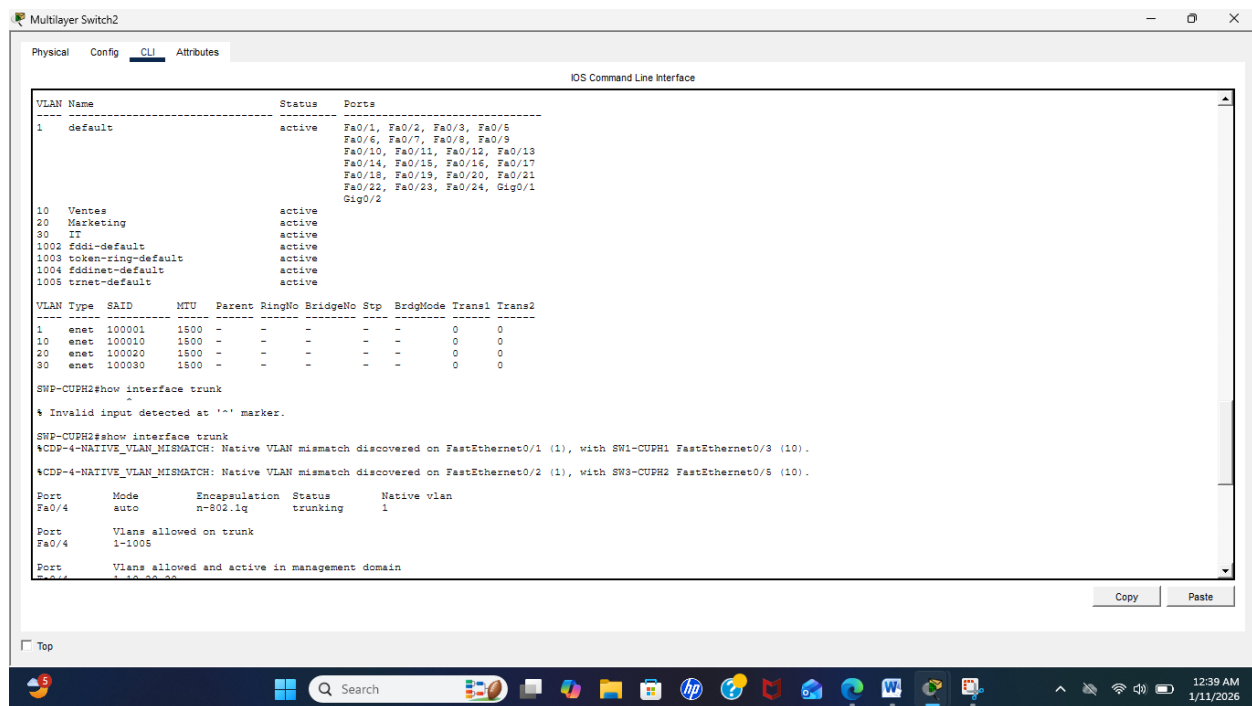
Switch# show vlan

b) Vérifier les trunks:

Switch# show interface trunk

c) Switch# show vtp status:

Switch# show spanning-tree summary



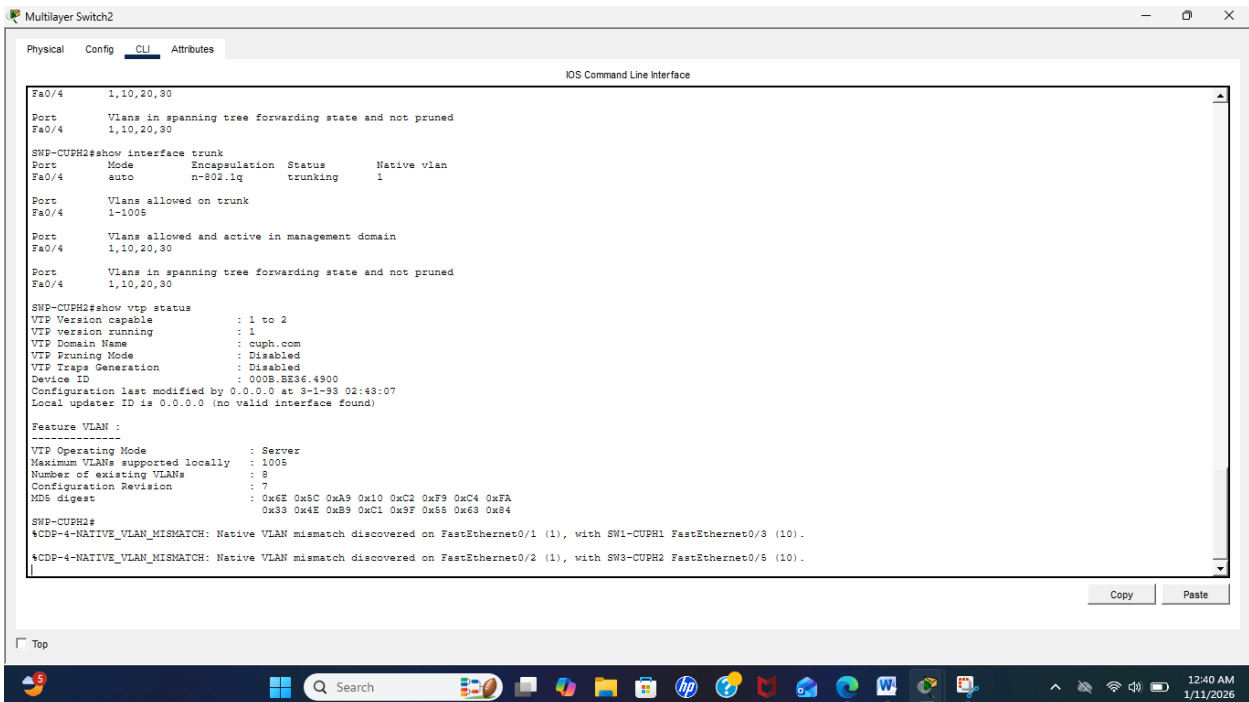


Figure 5 Sur cette image, j'ai fait des vérifications afin de montrer les VLANs, les trunks et le vtp status etc... (voit page 13 à 14).

9. Configuration IP des postes de travail

VLAN 10 (Ventes): 192.168.10.10/24 / Gateway: 192.168.10.1

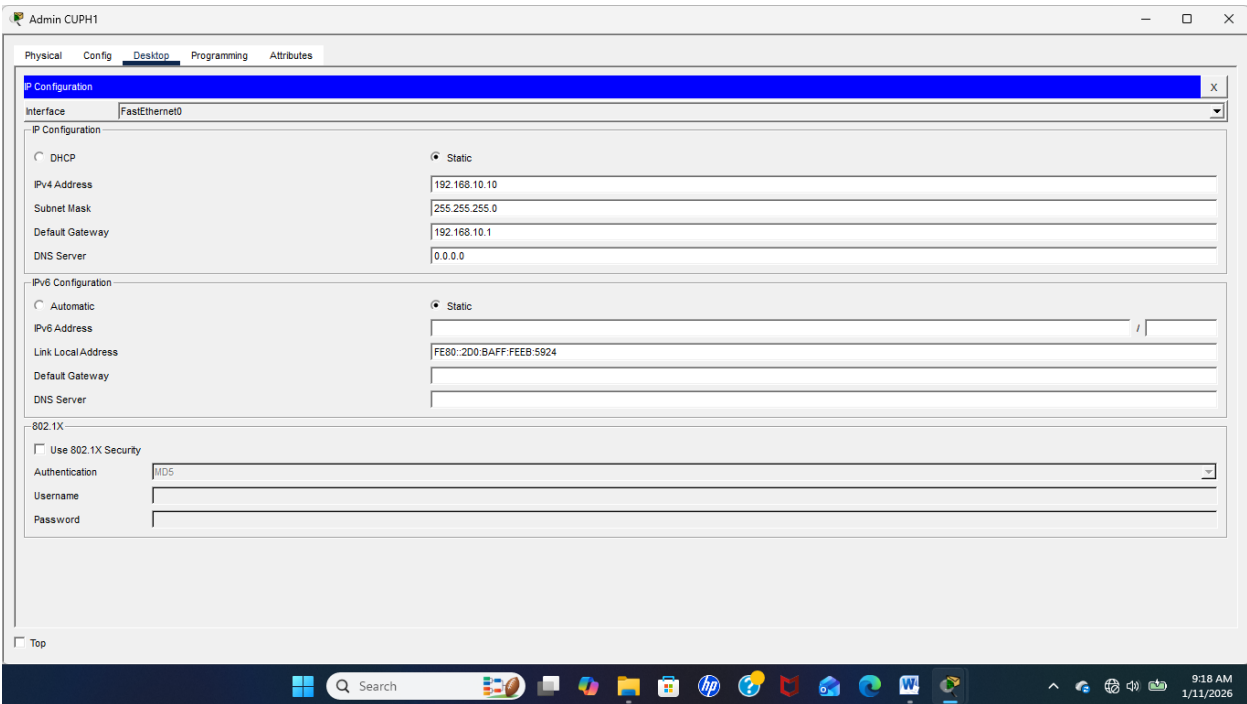
VLAN 20 (Marketing): 192.168.20.10/24 / Gateway: 192.168.20.1

VLAN 30 (IT): 192.168.30.10/24 / Gateway: 192.168.30.1

- **Réseau CUPH 1: Contenant deux PCs : Vlan 10 (Ventes)**

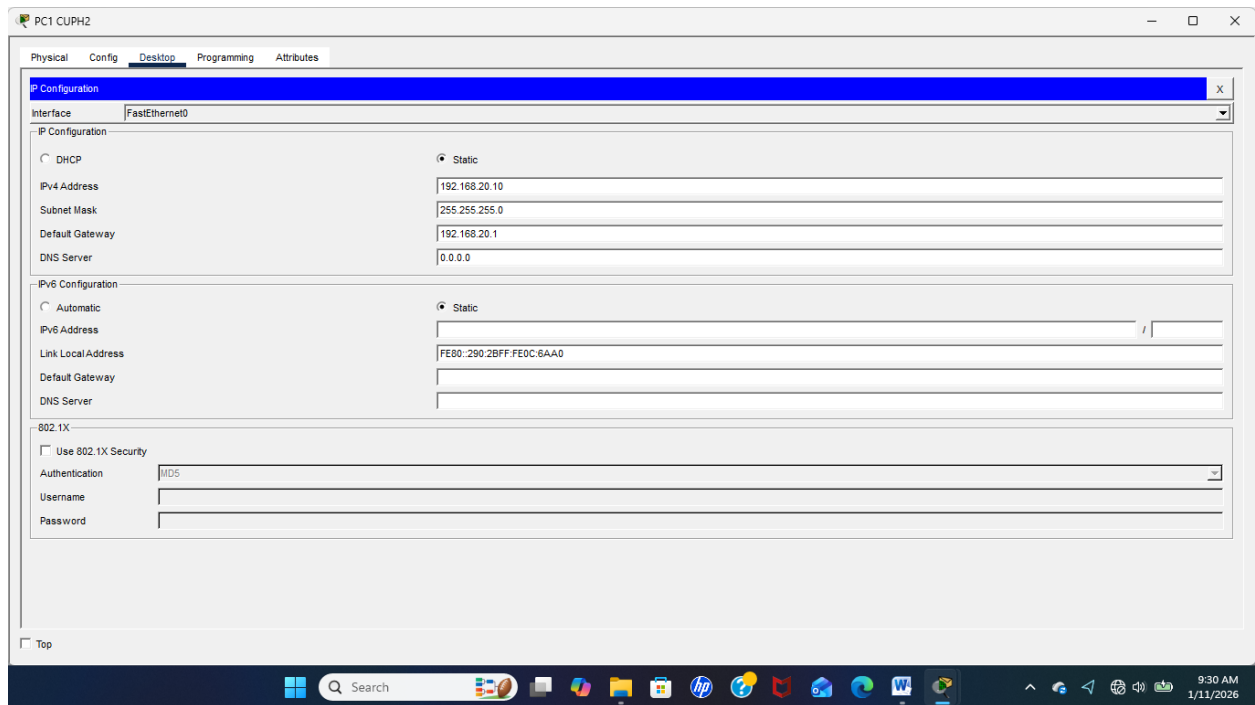
Admin- CUPH 1: IP: 192.168.10.10 Gateway: 192.168.10.1

PC- CUPH 1: IP: 192.168.10.11 Gateway: 192.168.10.1



- **Réseau CUPH 2: Contenant quatre (4) PCs, un PC- CUPH 1: Vlan 20 (Marketing)**

PC1- CUPH 2: IP: 192.168.20.10	Gateway: 192.168.20.1
PC2- CUPH 2: IP: 192.168.20.11	Gateway: 192.168.20.1
PC3- CUPH 2: IP: 192.168.20.12	Gateway: 192.168.20.1
PC4- CUPH 2: IP: 192.168.20.13	Gateway: 192.168.20.1



- **Servers et le Printer Vlan 30: IT**

Server Mail: 192.168.30.10	Gateway: 192.168.30.1
Server Web: 192.168.30.11	Gateway: 192.168.30.1
Server File: 192.168.30.12	Gateway: 192.168.30.1
Server DHCP: 192.168.30.13	Gateway: 192.168.30.1
Printer:	Gateway: 192.168.30.1

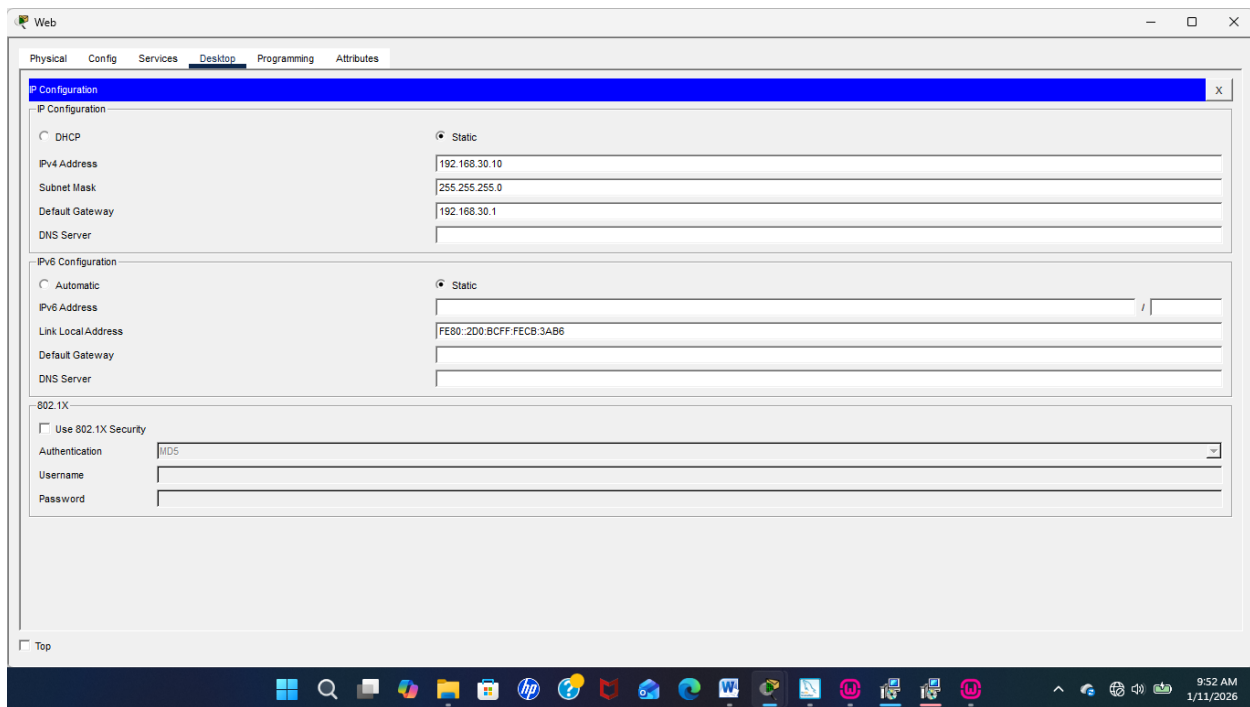


Figure 6 Sur cette image, j'ai distribué les addresses IP avec le Server Web

Nb: Cette exemple est considérer pour les autres dispositifs Vlan 30.

10- Tests de connectivité.

! Depuis un PC

ping 192.168.10.1

ping 192.168.20.1

- Un exemple sur le PC Admin

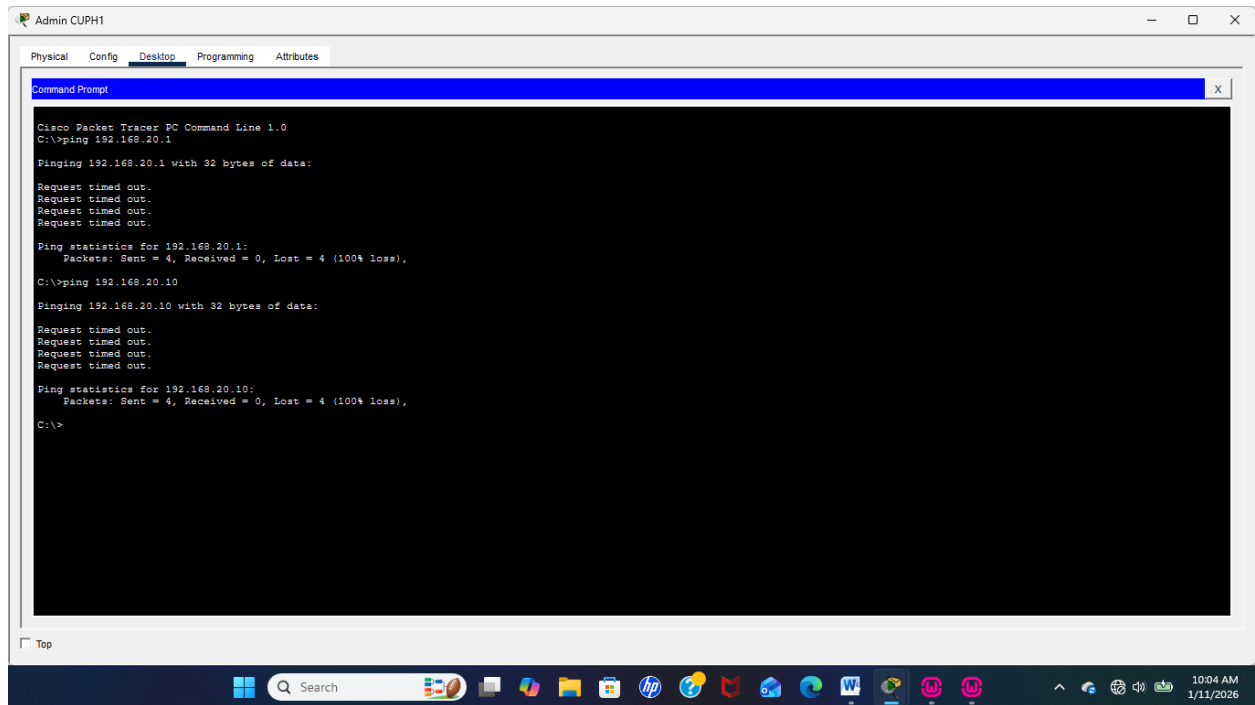
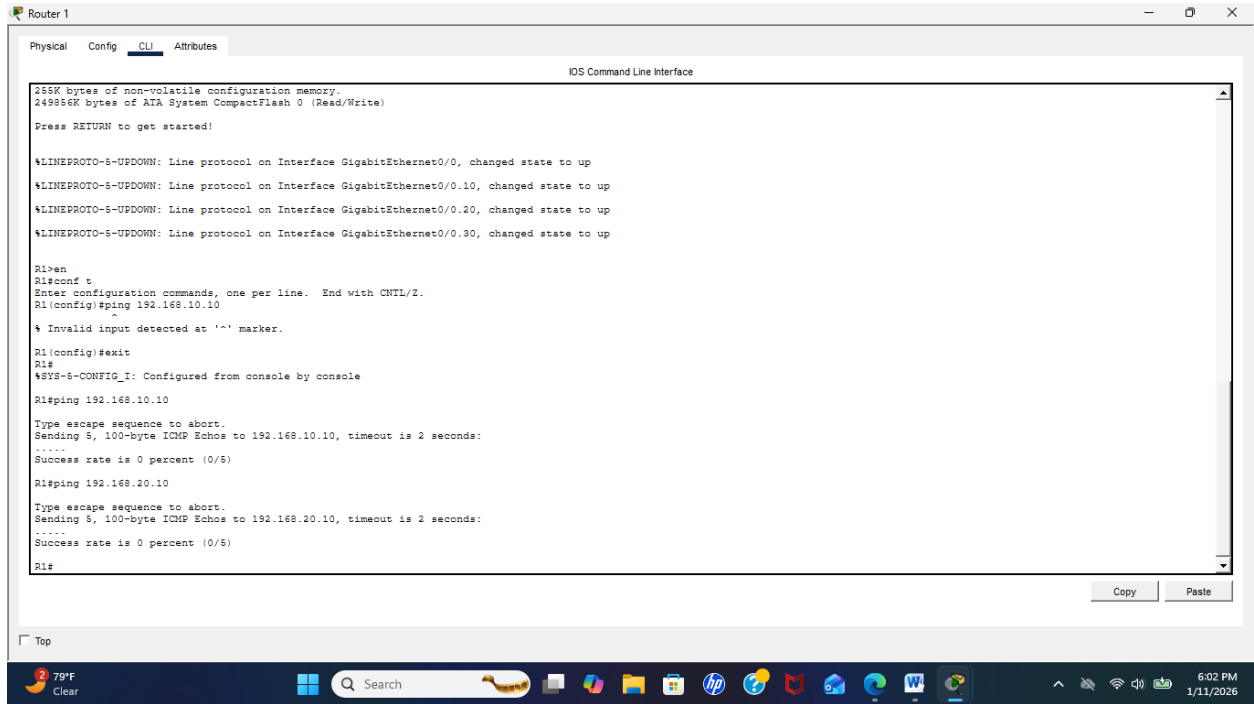


Figure 7 Sur le PC Admin j'ai épinglé deux PCs pas de communication, parce qu'il ont des base d'adresse différents

11- Depuis le routeur

ping 192.168.10.10

ping 192.168.20.10



```
Router 1
Physical Config CLI Attributes
IOS Command Line Interface

256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

R1>en
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ping 192.168.10.10

% Invalid input detected at '^' marker.

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#ping 192.168.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#ping 192.168.20.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echoes to 192.168.20.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#
```

- Sur cette image, j'ai effectué le test sur le routeur, donc la connexion est bien présente.

Conclusion,

Ce travail a permis de mettre en place une topologie réseau fonctionnelle basée sur les VLAN. La configuration des switchs et du routeur a été réalisée avec succès. Les tests de connectivité ont montré que les machines peuvent communiquer entre elles via le routage inter-VLAN, tout en maintenant une bonne organisation et une meilleure sécurité du réseau.